성질

- $\lim_{x\to a} f(x) = \alpha$

- $\lim_{x\to a} g(x) = \beta$

- $\lim_{x\to a} kf(x) = k\underbrace{\lim_{x\to a} f(x)}_{=\alpha} = k\alpha$

- $\lim_{x\to a} \{f(x) \pm g(x)\} = \alpha \pm \beta$

pf. $\underbrace{\lim_{x\to a} f(x)}_{=\alpha} \pm \underbrace{\lim_{x\to a} g(x)}_{=\beta}$

$\therefore \lim_{x\to a} \{f(x) \pm g(x)\} = \alpha \pm \beta$

- $\lim_{x\to a} f(x)g(x) = \alpha\beta.$

pf. $\underbrace{\lim_{x\to a} f(x)}_{=\alpha} \cdot \underbrace{\lim_{x\to a} g(x)}_{=\beta}$

$\therefore \lim_{x\to a} f(x)g(x) = \alpha\beta$

- $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\alpha}{\beta}$ $(\beta \neq 0,\ g(x) \neq 0)$

pf. $\dfrac{\lim_{x\to a} f(x) = \alpha}{\lim_{x\to a} g(x) = \beta}$

$\therefore \lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\alpha}{\beta}.$

etc)

- $f(x), g(x)$: 다항함수 AND $g(a) \neq 0$

$\lim_{x\to a} f(x) = f(a)$, $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{f(a)}{g(a)}$

- 대소관계.

When. $\lim_{x\to a} f(x) = \alpha$, $\lim_{x\to a} g(x) = \beta$

About $\forall x \to \alpha$

- $f(x) \leq g(x) \Rightarrow \alpha \leq \beta$

pf. $\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x) \Rightarrow \alpha \leq \beta.$

- $f(x) \leq h(x) \leq g(x)$ AND $\alpha = \beta.$

$\Rightarrow \lim_{x\to a} h(x) = \alpha$ or $\lim_{x\to a} h(x) = \beta.$

pf. $\underbrace{\lim_{x\to a} f(x)}_{=\alpha=\beta(\because \alpha=\beta)} \leq \lim_{x\to a} h(x) \leq \underbrace{\lim_{x\to a} g(x)}_{=\beta=\alpha(\because \alpha=\beta)}$

$\therefore \lim_{x\to a} h(x) = \alpha$